

## Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

**Thirty-two minutes across five segments** — three of them consecutive, from minute 8 to minute 30, and two late ones at 62 and 67.

**Conventions. ASK** — put it to the room before writing the next line, and the answer to expect.

**CHECK** — do it out loud; doing them is the habit being taught. **IF ASKED** — the question that always comes. **CUT** — drop this first if you are behind, decided now rather than at minute 70.

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## The board plan

Same room as session 3: blackboard across the front, screen down in the **centre**, a **left wing** and a **right wing**.

**LEFT WING — the ledger. Written once at minute 20, never wiped.** Two lines, and the session breaks without them:

$$v = \frac{V_{\max}[S]}{K_M + [S]}, \quad K_M \equiv \frac{k_{-1} + k_2}{k_1}, \quad V_{\max} \equiv k_2 E_{\text{tot}}$$

Every one of ConcepTest 1 (34 min), the two figures (38–46) and the handout debrief (58) points back at those definitions. The ConcepTest is unanswerable if they have been erased — the whole question is *what are these two symbols made of*.

**RIGHT WING — the workings, wiped between segments:** the nondimensionalisation, the four steps of the derivation, the two clocks, the  $81^{1/n}$  algebra, the four-state column.

**Two lines to carry forward.** At minute 30, under the ledger on the **left wing**, write and box both of these:

$$\frac{E_{\text{tot}}}{K_M + S_0} \ll 1 \quad \text{and} \quad E_{\text{tot}} \ll K_M + S_0$$

They are the same statement. The second is the projected slide at 30–34 min. The 42-minute figure is a plot of that same group, so the room needs to be looking at a symbol it derived rather than one that appeared on a slide.

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## 1 • 8–14 min — The small parameter

### Board 1 of 5 • 6 min • cumulative 6

From Thursday's cascade, already on their notes:

$$\frac{dm}{dt} = \alpha - \gamma_m m, \quad \frac{dp}{dt} = k_p m - \gamma_p p, \quad \gamma_m = 0.5, \gamma_p = 0.05$$

**ASK:** Which is the fast variable, and how do you know without simulating? Expect "mRNA, because  $\gamma_m$  is bigger." Push once: bigger **than what**? The answer you want is that  $\gamma_m/\gamma_p = 10$ , a pure number.

**Nondimensionalise in front of them.** Do not show the scaled equations finished — the point is watching the  $\varepsilon$  appear.

$$\tau = \gamma_p t, \quad \mu = \frac{m}{\alpha/\gamma_m}, \quad \pi = \frac{p}{k_p \alpha / \gamma_m \gamma_p}$$

Substitute. The mRNA equation becomes

$$\varepsilon \frac{d\mu}{d\tau} = 1 - \mu, \quad \frac{d\pi}{d\tau} = \mu - \pi, \quad \varepsilon \equiv \frac{\gamma_p}{\gamma_m} = 0.1$$

**Say it while the chalk is still on the board:**  $\varepsilon$  is the ratio of two clocks. It is not a statement about enzymes, or about concentrations, or about anything chemical.

**The sentence to land.** Setting  $\varepsilon = 0$  does not say  $dm/dt = 0$ . It says  $dm/dt$  is small compared with the terms in its own equation, each of which is order  $\alpha$ . Write both sides:

$$\frac{dm}{dt} = \underbrace{\alpha}_{\text{large}} - \underbrace{\gamma_m m}_{\text{large}} = \text{small}$$

**CHECK, out loud:** at  $\alpha = 10$ ,  $\gamma_m = 0.5$ : the two terms are each about 10 and their difference is about 1. Small because it is a difference of two big numbers, not because nothing is happening.

**The lost initial condition.** A first-order equation had one; the algebraic equation has none.  $m$  jumps to  $\alpha/\gamma_m$  instantly.

**IF ASKED** "so the approximation is just wrong at  $t = 0$ ?" — yes, and that is the honest answer. It is wrong for about  $1/\gamma_m \approx 1.4$  min. It is a boundary layer; they will meet the words in a fluids course. The error you cannot remove lives there.

**CUT if behind:** the boundary-layer paragraph. Keep the two-big-numbers check.

## 2 • 14–26 min — Michaelis–Menten, four named steps

Board 2 of 5 • 12 min • cumulative 18

Head the wing with the scheme:



**Ask for each step before you write it.** They can do 1 and 2 unaided; making them produce those two is what buys attention for 3 and 4.

**Step 1 — QSSA on the complex.**

***ASK:** Which is the fast variable here, and what replaces its differential equation? Expect ES. Then write:*

$$k_1[E][S] = (k_{-1} + k_2)[ES]$$

**Step 2 — enzyme conservation.**

***ASK:** What is conserved? Someone will say "the enzyme." Write it, then name it: this is a **left null vector** of the stoichiometric matrix, from Thursday. Thirty seconds, and it connects two things they hold separately.*

$$[E] + [ES] = E_{\text{tot}}$$

**Step 3 — eliminate [E].** Substitute and collect:

$$[ES] = \frac{E_{\text{tot}}[S]}{K_M + [S]}, \quad K_M \equiv \frac{k_{-1} + k_2}{k_1}$$

***Slow down here.**  $K_M$  is a **ratio of rate constants**. Half the room arrives believing  $K_M$  is "the concentration at half  $V_{\text{max}}$ " — which is a consequence of this definition, not the definition, and the difference bites the moment  $k_2$  is not small.*

**Step 4 — the rate.**

$$v = k_2[ES] = \frac{V_{\text{max}}[S]}{K_M + [S]}, \quad V_{\text{max}} \equiv k_2 E_{\text{tot}}$$

Now move the two definitions to the LEFT WING and leave them there.

**CHECK, out loud:** units.  $K_M = (k_{-1} + k_2)/k_1$  —  $s^{-1}$  over  $M^{-1}s^{-1}$  gives  $M$ . It is a concentration, and it is built entirely out of rate constants.

**IF ASKED** "isn't  $K_M$  just  $K_d$ ?" — only when  $k_2 \ll k_{-1}$ . Then  $K_M \rightarrow k_{-1}/k_1 = K_d$ . Write that limit down; the gap between  $K_M$  and  $K_d$  when it fails is a real experimental headache and it saves an argument in session 6.

**Close the segment on the debt, not on the result:** step 1 was an *assumption*, and nothing so far says when it is allowed.

**CUT if behind:** the  $K_M \rightarrow K_d$  limit. Nothing later depends on it.

### 3 • 26–30 min — Where the substrate goes

#### Board 3 of 5 • 4 min • cumulative 22

This segment pays the debt from segment 2. **It was rewritten on 7 September:** the earlier version derived the condition from  $\tau_1 < \tau_2$ , and that argument does not survive checking. Holding  $E_{\text{tot}}/(K_M + S_0)$  at 0.5 and driving  $\tau_1/\tau_2$  down three decades leaves the error pegged at 16%. Worse, the deck refuted itself — slide 9's right panel has  $\tau_1/\tau_2 = 0.023$ , clocks separated 44-fold, and Michaelis–Menten wrong by 15%. Clock separation is **necessary** and nowhere near sufficient. What controls the error is substrate bookkeeping.

**First: there is a fast variable.** One line each, no derivation needed.

$$\tau_1 \approx \frac{1}{k_1(E_{\text{tot}} + K_M + S_0)}, \quad \tau_2 \approx \frac{K_M + S_0}{V_{\text{max}}} = \frac{K_M + S_0}{k_2 E_{\text{tot}}}$$

Note the  $E_{\text{tot}}$  inside  $\tau_1$ . The older slide dropped it and claimed the fast clock was independent of enzyme; that is only true once you have already assumed the answer, and at  $E_{\text{tot}} = 1$  it is wrong by 46%.

**Say it and move on:** you need  $\tau_1 \ll \tau_2$  or there is no fast variable to eliminate. It is necessary. It is also nearly free, and on its own it buys you almost nothing.

**Second, and this is the one that bites: where does the substrate go?**

**ASK, and this is the question the segment exists for:** the reduced model has one equation, for  $S$ . The full model has  $S$  and  $ES$ . Where did the substrate in  $ES$  go? Expect a pause. The answer is that the reduced model has no account of it at all.

At quasi-steady state, and largest at the start:

$$[\text{ES}] = \frac{E_{\text{tot}} S}{K_M + S} \leq \frac{E_{\text{tot}} S_0}{K_M + S_0}$$

**Demand that be negligible against the substrate you are tracking:**

$$\frac{E_{\text{tot}} S_0}{K_M + S_0} \ll S_0 \implies \boxed{\frac{E_{\text{tot}}}{K_M + S_0} \ll 1} \quad \text{i.e.} \quad \boxed{E_{\text{tot}} \ll K_M + S_0}$$

One line. No bound, no bracket, no chain of inequalities — and it is the boxed line on the next slide, which is why that slide now says "nothing new here."

**CHECK, out loud:** the group is a fraction of substrate. That is what makes it dimensionless and that is why it is the thing the error scales with.

**The number that makes it physical, and it is on the next figure:** at  $E_{\text{tot}} = 1$ , 35% of all the substrate is sitting in complex at the peak. Michaelis–Menten is tracking the other 65% and calling it everything. That is the 15% error.

**IF ASKED** "so is the clock argument wrong?" — no, it is incomplete. You need both: a fast variable to eliminate (clocks) and a negligible amount of stuff hiding in it (bookkeeping). Only the second one sets the size of the error. Segel 1988 does both properly.

**Write both boxed lines on the left wing, under the ledger, and leave them.**

**CUT if behind:** the  $\tau_1/\tau_2$  lines. The bookkeeping argument stands alone and is the one the next three surfaces use.

## 4 • 62–67 min — What $n$ buys you

**Board 4 of 5 • 5 min • cumulative 27**

Four lines. This is **PS1 Q4b**, which asks for it analytically and forbids a numerical search — so what you write here is the worked method for a question they are about to be graded on.

$$\frac{x^n}{K^n + x^n} = 0.1 \implies x_{10} = K \left(\frac{1}{9}\right)^{1/n}, \quad \frac{x^n}{K^n + x^n} = 0.9 \implies x_{90} = K (9)^{1/n}$$

$$\frac{x_{90}}{x_{10}} = 81^{1/n}$$

**ASK:** where did  $K$  go? It cancels. The fold-change depends on  $n$  alone, which is why every curve in the projected family crosses half occupancy in the same place.

**The number to make them feel.**  $n = 1$  needs an **eighty-one-fold** change in input to go from a tenth on to nine tenths on. That is a slope, not a switch.  $n = 2$  needs nine-fold;  $n = 4$  needs three-fold.

**Then run it backwards, out loud — this is the engineering direction.** "My circuit has to switch on a three-fold change in input." Read the table the other way: three-fold means  $n \approx 4$ , so you need cooperativity, so you go looking for a protein that has it. Requirement  $\rightarrow$  number  $\rightarrow$  mechanism. That is the direction they will use in the project, and the table is a **spec sheet** read in that direction.

**IF ASKED** "is that why real switches are cooperative?" — yes, and it is the reason every toggle in the second half of this course has an  $n$  in it. Session 9's bistability condition is  $\alpha_c = n(n-1)^{-(n+1)/n}$ , which is **infinite** for  $n \leq 1$ : no cooperativity, no switch.

**CUT if behind:** the session-9 forward reference.

## 5 • 67–72 min — Two independent sites

Board 5 of 5 • 5 min • cumulative 32

**Write the four states as a column before you write  $Z$ .** This is a partition function, and it is session 6 arriving a week early — say so, because **PS1 Q5** asks the 247 students for exactly this and it would otherwise be assessing something untaught.

With  $w = x/K_d$ :

| STATE       | WEIGHT |
|-------------|--------|
| empty       | 1      |
| site 1 only | $w$    |
| site 2 only | $w$    |
| both        | $w^2$  |

$$Z = 1 + 2w + w^2 = (1 + w)^2$$

$$f_{\text{sites}} = \frac{\frac{1}{2}(w + w) + w^2}{(1 + w)^2} = \frac{w(1 + w)}{(1 + w)^2} = \frac{w}{1 + w}$$

**The square cancels.**  $n = 1$ , from two sites.

**MAKE THEM SAY WHICH FRACTION.** "Fraction of sites occupied" is  $w/(1 + w)$ . "Fraction of promoters with at least one site bound" is  $1 - 1/(1 + w)^2$ , which is **not a Hill function at all**. Same molecules, same physics, two different curves — and a paper may not tell you which one it plotted. This is the single most common mark lost on PS1 Q5.

**The generalisation, one line:**  $N$  independent identical sites give  $Z = (1 + w)^N$  and site occupancy  $w/(1 + w)$ , always. Independence is what makes the sites invisible to a dose–

response curve.

**ASK, to set up ConcepTest 2:** so what does a fitted  $n$  of 1.9 tell you? Leave it. That is the vote.

**The vote's answer is D, and you need both halves.** No ceiling:  $n \leq$  number of sites — that is the theorem you have just proved, and it is about an equilibrium **binding** curve. No floor: a **dose-response** is binding plus everything downstream, and downstream steps sharpen — zero-order ultrasensitivity or decoy titration gives an apparent  $n$  above 20 from a single site (both are session 13). B is right for the curve on this slide and wrong for the one in the stem. **Which curve did you measure?**

**CUT if behind:** the  $N$ -site generalisation. The two-site case carries the argument.